

TruckShield

System Documentation

360 Risk Partners

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1. System Overview

TruckShield is a comprehensive trucking insurance CRM and client portal built for 360 Risk Partners. The platform streamlines the entire insurance lifecycle from prospect intake to policy binding and renewal.

Production URL: truckshield.360riskpartners.com

User Roles

- **Staff (Admin):** Full system access including analytics, staff management, carrier configuration, and all account operations
- **Staff (Producer):** Account management, quoting, and client communication; scoped to assigned accounts for certain views. Producers can view and claim unassigned accounts.
- **Client:** Self-service portal for application completion, document uploads, messaging, and policy viewing

Technology

- React-based single-page application with TypeScript
- Lovable Cloud backend with row-level security policies
- Edge functions for server-side logic and integrations
- Transactional email infrastructure with queue-based delivery (pgmq)
- Real-time subscriptions for messaging and notifications
- Custom email domain (360riskpartners.com) for branded transactional and auth emails

2. Staff Dashboard Features

Account Management

- Create accounts via DOT number lookup (FMCSA/SAFER auto-fill) or manual entry
- Duplicate DOT detection prevents creating duplicate accounts
- Search and filter accounts by company name, status, or assigned producer
- Each account tracks company details, contact info, fleet data, and application progress

Pipeline View

A drag-and-drop Kanban board visualizes accounts across the insurance lifecycle:

Pipeline Column	Description
Pending Info	Account created; awaiting client application completion
Info Complete	Client has submitted the 10-step application
Quoting	At least one carrier submission has been made
Quoted	A quote has been received from a carrier
Bound	Coverage has been bound with final premium and policy documents

- Stale account warnings highlight accounts inactive for extended periods
- Filter by assigned producer to focus on individual workloads

Account Detail View

- Full account record with collapsible sections for easy navigation
- Producer assignment (admin-only) to delegate account ownership
- Status management with manual override capability
- Application preview showing all 10 steps of client-submitted data
- PDF export of the complete application for carrier submissions

Carrier Manager

- Add and edit carriers with detailed appetite criteria
- Criteria include: fleet size range, revenue range, authority age, cargo types, operating states, business types
- Upload carrier appetite guides (PDF) for AI-powered analysis
- Upload carrier logos for display in client-facing quote views

AI Market Guidance

- Gemini-powered carrier matching analyzes account data against carrier appetite
- Scores each carrier 0–100 with tiers: Strong Match, Partial Match, Poor Match
- Provides specific strengths and concerns for each carrier match

- “Mark as Submitted” action creates a quote record and moves the account into the quoting pipeline

Submitted Markets

Track carrier quote statuses through the quoting lifecycle:

Status	Description
Submitted	Application sent to carrier
Under Review	Carrier is reviewing the submission
Info Requested	Carrier needs additional information from the client
Quoted	Carrier has provided a premium quote
Declined	Carrier has declined coverage (reason required)
Bound	Coverage bound with final premium and policy document

- Upload quote documents and bind documents
- Update quotes with revised premium amounts
- Request additional info from client (triggers email notification)

Document Hub

- Categorized file storage: Loss Runs, Cab Cards, Quotes, Policies, MVR, Drivers License, Misc
- Upload with category selection and automatic standardized naming
- Naming convention: [Category]_[CompanyName]_[Date].ext

Task Manager

- Create tasks per account with priority levels (Low, Medium, High)
- Assign due dates and team member ownership
- Track completion status

Activity Log

- Chronological audit trail of all actions performed on an account
- Includes status changes, document uploads, quote updates, and communications

Messaging

- Real-time bidirectional chat per account between staff and client
- File attachment support
- Read receipts indicate when messages have been viewed

Analytics Dashboard (Admin-only)

- Pipeline funnel chart showing account distribution across statuses
- Status distribution breakdown

- Stale account alerts for accounts requiring attention
- Producer performance metrics

Staff Management (Admin-only)

- View all staff members and their roles
- Invite new staff via email with role assignment (admin or producer)

PDF Import

- Upload existing PDF applications for AI-powered data extraction
- Extracted data populates account fields automatically

Client Invite

- Send or resend single-click magic link invitations to clients
- Auto-triggered when contact email is set on an account

3. Client Portal Features

Journey Timeline

Visual 4-stage progress indicator showing the client where they are in the process:

Submission → Quoting → Quoted → Bound

Application Wizard

A 10-step guided form for complete application submission:

Step	Section	Key Fields
1	Applicant Info	Company details, contact info, DOT/MC numbers
2	Coverage Selections	Types of coverage needed, effective dates
3	Radius of Operations	Operating radius percentages by distance tier
4	Commodities	Types of cargo hauled with percentages
5	Power Units	Trucks with VIN, year, make, model, values
6	Trailers	Trailer details including non-owned trailers
7	Drivers	Driver info, license details, violations, accidents
8	Loss History	Prior coverage history by coverage type
9	General Questions	Operational and safety questions
10	Review & Submit	Review all sections, upload documents, submit

- Auto-save preserves progress across sessions
- Red border validation highlights incomplete required fields
- Inline cab card and loss run document uploads

Info Request Banner

- Persistent alert displayed when a carrier has requested additional information
- Links directly to the relevant section for the client to respond

Quote Display

- Shows carrier logo, premium amount, and downloadable quote PDF
- Only visible after a carrier has provided a quote

Policy Renewal Card

- Displayed for bound policies showing policy expiration date and countdown
- Helps clients track when renewal action is needed

Document Hub

- View and download documents shared by staff

- Upload additional documents as requested

Messaging

- Chat with agency staff with file attachment support

4. Authentication & Onboarding

Client Authentication — Single-Click Magic Link

Clients use passwordless authentication. The invitation flow has been streamlined into a single-click experience:

- Staff invites a client → the system generates a pre-authenticated magic link via the create-client-magic-link edge function
- The magic link combines Supabase authentication with the invitation token in one URL
- Client clicks the link in their email → authenticated and redirected to portal instantly (one step)
- No email/password entry, no second magic link email required

Fallback for expired links:

- If the magic link has expired (default 1 hour), the Auth page detects the invite token and pre-fills the client's email
- A "Send me a new link" button lets the client request a fresh magic link with one click
- Clear error banners for expired, invalid, or already-accepted invitations

Previous flow (deprecated): Invite email → Click link → Type email → Wait for second magic link email → Click link → Portal

Current flow: Invite email → Click link → Portal

Staff Authentication

- Email/password signup via invitation link
- Optional magic link sign-in for staff
- Token-based role assignment ensures proper access levels
- `accept_staff_invitation` function validates token and grants appropriate role (admin/producer)
- Password reset via email with dedicated `/reset-password` page

Client Invitation Auto-trigger

Whenever a `contact_email` is set on an account (via the application wizard, SAFER import, or manual entry), an invitation is automatically sent to the client with a single-click magic link.

Staff Invitation Flow

- Admin generates a time-limited invitation token
- Email sent to the new staff member with signup link
- Staff member signs up or logs in via the invite link
- `accept_staff_invitation` function grants the appropriate role (admin/producer)

5. Automated Status Progression

The `auto_update_account_status` database trigger automatically transitions account status based on system events:

Trigger Event	From Status	To Status
Application completed (step $\geq$ 10)	pending_info	info_complete
First quote submitted to carrier	pending_info / info_complete	quoting
Quote received from carrier	quoting	quoted
Coverage bound	any	bound

Each transition automatically logs an activity entry and fires the `notify-status-change` edge function for email notifications to the client and assigned producer.

6. Automated Emails (11 Templates)

All outgoing client emails are automatically CC'd to the assigned producer for the account (if one is assigned). The producer receives a duplicate of every client-facing email.

Template	Trigger	Recipient	Contents
Client Portal Invite	Contact email set on account	Client	Single-click magic link to access portal
Application Received	Application submitted (status → info_complete)	Client	Confirmation of application receipt
Application Completed (Staff)	Application submitted	All admins + assigned producer	Company name, DOT#, submitter name
Pipeline Status Change	Account status changes (quoting/quoted/bound)	Client	Status update with portal link
Carrier Status Change	Carrier quote status updated	Client	Carrier name, new status, portal link
Additional Info Request	Staff requests info from carrier panel	Client	Carrier name, request details, portal link
Application Reminder	Daily cron (every 3 days)	Clients with incomplete apps	Completion percentage, portal link
Application Not Started	Daily cron (every 3 days)	Clients who haven't begun application	Prompt to start application, portal link
Info Request Reminder	Daily cron (every 3 days)	Clients with pending info requests	Carrier name, details, days pending
Invite Reminder	Daily cron (every 3 days)	Clients who haven't accepted invite	Reminder to access portal, portal link
New Message Received	New message sent	Opposite party (client ↔ staff)	Message preview, portal link

Carrier Status Change covers these statuses: Submitted, Under Review, Quoted, Quote Updated, Declined, Bound.

Producer CC behavior: When an email is sent to a client, the system looks up the account's assigned producer, retrieves their email from the profiles table, and sends them a duplicate email with a unique idempotency key. The CC is skipped if no producer is assigned, if the producer's email matches the client's, or if the producer is suppressed/unsubscribed.

7. In-App Notifications

The `create_notification_on_event` database trigger creates bell icon notifications for:

- **New message from client:** All admins + assigned producer receive notification
- **New message from staff:** Client receives notification
- **Info requested by carrier:** Client receives notification
- **Quote available:** Client receives notification

8. Scheduled Jobs

Reminder Emails Cron

- Runs daily via cron schedule
- Application reminders: sent to clients with incomplete applications in `pending_info` status who have a linked client user
- Application not started: sent to clients who were invited but have not begun the application (step 0 or null)
- Info request reminders: sent to clients with pending (unresolved) info requests
- Invite reminders: sent to clients who have a pending invitation that hasn't been accepted yet
- Throttled to every 3 days per recipient to avoid email fatigue
- Producer CC: All reminder emails are also sent to the assigned producer

Email Queue Processing

- `process-email-queue` cron runs every 5 seconds via `pg_cron`
- Processes `auth_emails` queue first (high priority), then `transactional_emails` queue
- Automatic retry on transient failures with dead-letter queue after 5 attempts
- Rate-limit handling with `Retry-After` backoff

9. Document Naming Convention

All file uploads follow a standardized naming convention:

`[Category]_[Company_Name]_[Suffix]_[Date].[ext]`

Examples

- `Loss_Run_Acme_Trucking_2026-04-07.pdf`
- `Quote_Acme_Trucking_Cover_Whale_2026-04-07.pdf`
- `Policy_Acme_Trucking_National_Indemnity_2026-04-07.pdf`

Applied to: Loss runs, cab cards, Document Hub uploads, quote uploads, and bind documents.

10. Key Workflows

New Account to Bound Policy

The end-to-end workflow from account creation to bound coverage:

- DOT Lookup (SAFER) → Account Created → Status: Pending Info
- Client invitation auto-sent with single-click magic link when contact email is set
- Client clicks link → authenticated instantly → completes 10-step application → Status: Info Complete
- Staff runs AI Market Guidance for carrier matching
- Staff marks carriers as Submitted → Status: Quoting
- Carriers review and provide quotes → Status: Quoted
- Staff enters final premium, uploads binder → Status: Bound
- Policy Renewal Card displayed to client with expiration countdown

Pipeline Flow

Pending Info → Info Complete → Quoting → Quoted → Bound

Client Onboarding Flow

Single-click magic link eliminates multi-step signup friction:

- Staff sets contact email on account (manual, SAFER, or application wizard)
- System generates a pre-authenticated magic link via create-client-magic-link edge function
- Invite email sent with the magic link as the CTA button
- Client clicks link → Supabase verifies OTP → session created → invitation accepted → portal displayed
- If link expired: Auth page shows pre-filled email + “Send me a new link” button

11. Edge Functions Reference

Function	Purpose
create-client-magic-link	Generates pre-authenticated magic links for client invitations using Supabase Admin API
fmcsa-lookup	Queries FMCSA/SAFER database for DOT number data
ai-market-guidance	Gemini-powered carrier matching and scoring
extract-appetite-guide	AI extraction of carrier appetite criteria from PDF uploads
extract-pdf-application	AI extraction of application data from uploaded PDF forms
coverwhale-api	Integration with Cover Whale API for automated quoting
ctq-webhook	Webhook handler for Cover Whale quote responses
notify-status-change	Sends pipeline and carrier status change emails to client + producer
send-transactional-email	Central email sender for all transactional email templates
send-reminder-emails	Daily cron-triggered reminder emails (application, info request, invite, not started)
send-notification-email	Legacy notification email sender
auth-email-hook	Custom branded auth email templates (signup, recovery, magic link, etc.)
process-email-queue	Queue dispatcher for email delivery with retry and rate-limit handling
handle-email-unsubscribe	Validates and processes email unsubscribe tokens
handle-email-suppression	Webhook handler for email bounces and complaints
preview-transactional-email	Renders email template previews for the dashboard

12. Security Architecture

- Row-level security (RLS) on all database tables
- Roles stored in separate user_roles table (never on profiles) to prevent privilege escalation
- guard_client_account_updates trigger prevents non-staff from modifying protected fields (status, producer assignment, created_by, client_user_id)
- Producers can only assign themselves to unassigned accounts; cannot reassign accounts owned by others
- Storage buckets use path-based policies enforcing account_id/ prefixes
- Client invitation visibility restricted by matching recipient's email against auth JWT claim
- Notification write access reserved for system-level triggers via service role
- Edge functions validate JWT claims in code (getClaims) for authenticated endpoints
- Email suppression system prevents sending to bounced/complained/unsubscribed addresses

Version History

Version	Date	Changes
v1.0	March 2026	Initial documentation
v2.0	March 2026	Added invite reminder, application not started templates, document naming, PDF import
v3.0	April 2026	Single-click magic link invites, producer CC on all client emails, domain migration to 360riskpartners.com, edge functions reference, security architecture section, email queue documentation, updated to 11 email templates